Canvas Quiz 5, Modules 7b–8 (graded)

Module 7b - PK-PD Relationship · Canvas quiz · 19 Oct 2026

Name _____ Date _____

Question 1 For a drug that displays linear, first-order pharmacokinetics, what is the impact on duration of activity if there is an increase in elimination half-life?

A. Duration of activity will increase.

B. There is a no change to the duration of activity.

C. Duration of activity will decrease.

Question 2 Which of the following results when the IV bolus dose of a drug displaying linear, first-order pharmacokinetics is doubled?

A. The half-life is decreased by half

B. The duration of action is increased by one half-life

C. The duration of action is doubled

D. The half-life is doubled

Question 3 What is the duration of action following a 604 mg IV bolus dose of a drug that displays linear, first-order pharmacokinetics? The half-life of elimination is 5 hours, apparent volume of distribution is 18 L, and minimum effective concentration is 6 mcg/mL. Provide your answer in hours rounded to the nearest tenth (one decimal place) with no units.

Question 4 What dose of a medication, that is to be administered as a single IV bolus injections would provide a duration of action of 20 hours in a 75 kg patient? The drug displays linear, first-order pharmacokinetics, has an elimination half-life of 5.8 hours, apparent volume of distribution of 0.25 L/kg, and a minimum effective concentration of 4 mg/L. Provide your answer in mg rounded to the nearest whole number with no units.

Question 5 Which of the following is true for a medication that displays nonlinear pharmacokinetics?

A. At low concentrations of drug in the plasma relative to Vmax, the elimination rate is independent of concentration.
B. At low concentrations of drug in the plasma relative to KM, the elimination rate depends on the concentration.
C. At low concentrations of drug in the plasma relative to Vmax, the elimination rate depends on concentration.
D. At low concentrations of drug in the plasma relative to KM, the elimination rate is independent of concentration.

Question 6 Which of the following is true regarding drugs that display nonlinear pharmacokinetics?

A. The AUC is proportional to the amount of bioavailable drug.
B. Half-life changes as dose is increased
C. Parameters do not change when different or multiple doses of a drug are given
D. The principle of superposition applies at steady-state.

Question 7 What daily dose rate of a non-linear drug would provide a steady-state concentration of 15 mg/L? In this patient, the KM is 7.9 mg/L and the Vmax is 625 mg/day. Provide your answer in mg/day rounded to the nearest five with no units.

Question 8 When a 25 year-old, 75 kg patient was taking a daily oral dose of 350 mg, his steady state plasma concentration was 8.7 mg/L.When the daily dose was increased to 450 mg, a steady-state plasma concentration of 20.6 mg/L resulted. What is the KM of this drug in this patient? Provide your answer in mg/L rounded to the nearest tenth (one decimal place) with no units.